

Math 421

Monday, October 13

Announcements

- Exam 1
 - Wednesday 10/15 – 5:45-7:15pm
- This week's drop-in hours
 - **Tuesday** 9-11 am VV B205
 - Wednesday 12-1 pm **VV B205**
- No homework due Friday 10/17

Chapter 2: Sequences

Section 10 (Part 1): Monotone Sequences

Monotonic Sequences

Definition. A sequence (s_n) of real numbers is called **monotonic** if it is increasing or decreasing.

- A sequence (s_n) of real numbers is called an **increasing sequence** if $s_n \leq s_{n+1}$ for all n .
- A sequence (s_n) of real numbers is called a **decreasing sequence** if $s_n \geq s_{n+1}$ for all n .

Activity – Give an example of the following

- An increasing sequence
- A decreasing sequence
- A non-monotonic sequence.

True/False

- ❖ Every convergent sequence is bounded.
- ❖ Every bounded sequence is convergent.
- ❖ Every convergent sequence is monotonic.
- ❖ Every monotonic sequence is convergent.

Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. *All bounded monotone sequences converge.*

Discussion.

Bounded Monotonic $\Rightarrow$ Convergent

Theorem 10.2. *All bounded monotone sequences converge.*

Proof.

Unbounded Monotonic $\Rightarrow$ Diverges to Infinity

Theorem 10.4.

- i. If (s_n) is an unbounded increasing sequence, then $\lim s_n = +\infty$.
- ii. If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.

Proof of i.

Chapter 2: Sequences

Section 11 (Part 1): Subsequences

Subsequence

Definition. A **subsequence**, (t_n) , of a sequence, (s_n) , is a new sequence formed by selecting, in order, some (possibly all) of the terms in (s_n) .

Examples.

Let (s_n) be the sequence defined by $s_n = n^2(-1)^n$.

Activity

1. Let $a_n = 3 + 2(-1)^n$ for $n \in \mathbb{N}$.

- List the first four terms of the sequence (a_n) .
- What terms could you select to produce a subsequence that is constant – i.e. the set of values is a single element?

2. Let $b_n = (-1)^n$ for $n \in \mathbb{N}$.

- What terms could you select to produce a subsequence that contains only positive terms?
- What terms could you select to produce the following subsequence $(1, 1, -1, -1, 1, 1, -1, -1, \dots)$?

Theorem 11.2

Thm. Let (s_n) be a sequence.

1. If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
2. If the sequence (s_n) is unbounded above, it has a subsequence with limit $+\infty$.
3. If the sequence (s_n) is unbounded below, it has a subsequence with limit $-\infty$.

In all cases, the subsequence can be taken to be monotonic.

Discussion Questions

- ❖ What is part 1 of the theorem saying you your own words?
- ❖ Convince yourselves the forward direction of part 1 is true: If there is a subsequence of (s_n) converging to t then set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
- ❖ Convince yourselves parts 2 and 3 are true.

Theorem 11.2 – Proof of 1

Discussion.

Theorem 11.2 – Proof of 1

Discussion. (cont.)

Example – Sequence of rational numbers

Theorems involving subsequences

Theorem 11.3: If the sequence (s_n) converges, then every subsequence converges to the same limit.

Theorem 11.4: Every sequence (s_n) has a monotonic subsequence.

❖ Convince yourselves these statements are true.

Bolzano-Weierstrass Theorem

Thm: Every bounded sequence has a convergent subsequence.

❖ Why is this theorem true?

Subsequential Limit

Def. Let (s_n) be a sequence in $\mathbb{R}$. A **subsequential limit** is any real number or symbol $+\infty$ or $-\infty$ that is the limit of some subsequence of (s_n) .

Example.